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10.1029/2024JD042809

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Synergy of Multiple-Satellite Measurements to Fill the Gap of Global XCO_2

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Abstract Accurate monitoring of atmospheric carbon dioxide (CO_2) concentrations is essential for understanding the global carbon cycle. Satellite remote sensing enables global-scale CO_2 monitoring, providing column-averaged dry air mole fractions of CO_2 (XCO_2). Current missions such as the Orbiting Carbon Observatory-2 (OCO-2) and Greenhouse gases Observing SATellite (GOSAT) series provide high-quality XCO_2 but have many observation gaps because of narrow swath widths or cloud interference. To address this, we first propose a machine learning (ML) approach to estimate daily XCO_2 (ML XCO_2) at a 0.25° resolution using OCO-2, Sentinel-5 Precursor TROPOspheric Monitoring Instrument, and ERA5 reanalysis data from May 2018 to December 2023. Validation against the Total Carbon Column Observing Network measurements shows high agreement, with an R^2 of 0.95, a root mean square error of 1.05 ppm, and a mean absolute error of 0.78 ppm. The spatiotemporal variations in ML XCO_2 are generally consistent with OCO-2, GOSAT, and the Copernicus Atmospheric Monitoring Service (CAMS) XCO_2. Notably, during the Coronavirus disease 2019 (COVID-19) pandemic in 2020, ML XCO_2 estimates were more consistent with OCO-2 and GOSAT XCO_2 than CAMS XCO_2, which was overestimated owing to unadjusted COVID-19-related CO_2 emission reductions. The annual mean CO_2 growth rates from ML XCO_2 (2.065–2.735 ppm/year, 2019–2023) also agree with estimates from OCO-2 and the National Oceanic and Atmospheric Administration surface measurements, indicating the robustness of our approach. Our study demonstrates that the synergy between multiple-satellite measurements enhances the spatial coverage of XCO_2 and improves our understanding of the global carbon cycle.

Plain Language Summary Carbon dioxide (CO_2) is the most important greenhouse gas that drives global warming and climate change. Monitoring atmospheric CO_2 levels is crucial for tracking progress in reducing CO_2 emissions. Satellite remote sensing plays an important role in measuring CO_2 levels globally, but current operating satellites provide sparse CO_2 measurements limiting continuous CO_2 monitoring. To address this, this work proposes a machine learning approach to estimate global CO_2 levels by combining data from multiple satellites. Our estimates showed good agreement with ground-based measurements distributed globally, confirming the robustness and accuracy of our approach. During the COVID-19 pandemic, our estimates were more consistent with other satellite CO_2 products than model-based products, which were overestimated during this period. Additionally, the growth rates of atmospheric CO_2 from our estimates were generally consistent with both satellite and ground-based measurements. By combining data from multiple satellites, our machine learning approach enhances the spatial coverage of CO_2 data with reliable accuracy, contributing to a better understanding of the global carbon cycle.

Key Points:

- We propose a machine learning approach to estimate daily 0.25° column-averaged dry air mole fractions of carbon dioxide (XCO_2) on a global scale
- Our approach successfully filled the observation gaps in XCO_2 using multiple-satellite measurements with reliable accuracy
- Our results offer valuable insights to enhance the understanding of the global carbon cycle

Supporting Information:

Supporting Information may be found in the online version of this article.

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Citation:

Lee, J., Jeong, S., Kim, Y. J., Roh, S., Kim, J., & Jin, H. (2025). Synergy of multiple-satellite measurements to fill the gap of global XCO_2. *Journal of Geophysical Research: Atmospheres*, *130*, e2024JD042809. https://doi.org/10.1029/2024JD042809

Received 26 OCT 2024
Accepted 3 AUG 2025

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1. Introduction

Carbon dioxide (CO_2) is the most significant greenhouse gas contributing to global warming and climate change (IPCC, 2007). Anthropogenic activities, notably the use of fossil fuels, have dramatically increased global atmospheric CO_2 concentrations from approximately 278 parts per million (ppm) during the Industrial Revolution to 419.31 ± 0.15 ppm in 2023 (Lan et al., 2024). More than half of this increase has occurred since 1985, and over a quarter has occurred since 2000 (Crisp et al., 2022). This rise in CO_2 levels has intensified the frequency and impact of extreme weather events, such as heat waves, heavy rainfall, droughts, and heavy snowfalls (Clarke et al., 2022). In response, the international community adopted the Paris Agreement in 2015, aiming to limit

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Writing – review & editing: Jonghyuk Lee, Sujong Jeong, Soona Roh, Jiyeon Kim, Hyungah Jin

global warming to well below 2°C compared to preindustrial levels. Additionally, the Intergovernmental Panel on Climate Change (IPCC) approved the Special Report on Global Warming of 1.5°C in 2018 urging a reduction in global CO_2 emission by over 45% by 2030 and achieving carbon neutrality by 2050 (IPCC, 2018). Accurate and continuous monitoring of atmospheric CO_2 on a global scale is essential for effective mitigation of CO_2 emissions.

Satellite observations are crucial for tracking atmospheric CO_2 globally. These observations primarily provide the column-averaged dry air mole fractions of CO_2 (XCO_2), which are derived by analyzing the high-resolution spectra of sunlight reflected by the Earth's surface within the oxygen (O_2) and CO_2 absorption bands. The SCanning Imaging Absorption spectroMeter for Atmospheric CHartographY (SCIAMACHY) onboard the Environmental Satellite (ENVISAT) was the first instrument to provide global XCO_2 operating from 2002 to 2012 (Burrows et al., 1995). This instrument provides XCO_2 data with a spatial resolution of 30×60 km^2, pioneering high-resolution CO_2 data from satellites. In 2009, the Japanese Greenhouse gases Observing SATellite (GOSAT), the first satellite dedicated to greenhouse gas observation, was launched (Kuze et al., 2009). The GOSAT satellite is equipped with a Fourier Transform Spectrometer (FTS) that observes atmospheric greenhouse gases, including CO_2 and methane (CH_4), enhancing the spatial resolution to approximately 10.5 km with a temporal resolution of 3 days. Following this, the National Aeronautics and Space Administration (NASA) Orbiting Carbon Observatory-2 (OCO-2) was launched in 2014, which was the first NASA's satellite dedicated to observing atmospheric CO_2. OCO-2 provides global XCO_2 data at an even finer spatial resolution of approximately 1.29×2.25 km^2 and a 16-day temporal resolution (Crisp et al., 2017; Eldering, O'Dell, et al., 2017). Building on the success of these missions, subsequent satellites such as GOSAT-2 (Imasu et al., 2023) and OCO-3 (Taylor et al., 2020) have been launched to continue and expand the capability of global CO_2 monitoring. Additionally, the Chinese Global Carbon Dioxide Observation Satellite Mission (TanSat), which operated from 2016 to 2020, has also made significant contributions to global CO_2 monitoring (Y. Liu et al., 2018). These satellites provide highly accurate global XCO_2 data that have been utilized in various research applications, such as quantifying CO_2 emissions (Nassar et al., 2017, 2021), evaluating emission inventories (Yang et al., 2020), estimating CO_2 flux at the scale of megacities or larger regions (Hakkarainen et al., 2016; Janardanan et al., 2016; Kort et al., 2012), and estimating terrestrial carbon sinks (L. Zhang, Jiang et al., 2023).

However, despite their effectiveness, currently operating CO_2 monitoring satellites have limitations owing to many observation gaps. For instance, the OCO-2 satellite has a swath width of approximately 10 km and revisit every 16 days, resulting in large gaps in data coverage. This is because OCO-2 was designed as a sampling mission and not as a mapping mission (Eldering, Wennberg, et al., 2017). In comparison, the GOSAT and GOSAT-2 satellites revisit every three and six days, respectively. Although these revisit cycles are better than those of OCO-2, their discrete sampling strategy at several observation grid points and a relatively large spatial resolution of approximately 10 km result in fewer data sets compared to OCO-2 (Crisp et al., 2022). Moreover, atmospheric conditions, such as clouds and high aerosol levels, can further obstruct CO_2 observations, leading to additional gaps in satellite-based CO_2 measurements (J. Hong et al., 2023). These observation gaps present significant challenges for the continuous monitoring of CO_2 and accurate assessment of the global carbon cycle.

Numerous approaches have been developed to fill gaps in satellite-based CO_2 observations, including data fusion-based methods (Y. Wang et al., 2023) and geostatistical interpolation-based methods (or kriging-based methods) (Bhattacharjee & Chen, 2020; Sheng et al., 2023). Recently, many researchers have applied machine learning (ML) and deep learning approaches to fill in satellite-based XCO_2 products (C. He et al., 2022; Q. He et al., 2023; S. He et al., 2023; T. Li et al., 2023; Siabi et al., 2019; W. Wang et al., 2022; C. Wu et al., 2023; M. Zhang & Liu, 2023; L. Zhang et al., 2022). For example, Siabi et al. (2019) used an artificial neural network (ANN) model to derive the spatial distribution of XCO_2 in Iran. C. He et al. (2022) derived the full-coverage spatiotemporal distribution of daily XCO_2 across China from 2015 to 2018 using a Light Gradient Boosting Machine (LightGBM) model (Ke et al., 2017). C. Wu et al. (2023) used a spatial Categorical Boosting (CatBoost) model to reconstruct the XCO_2 across China from 2012 to 2019 at a resolution of 1 km. These approaches successfully fill the XCO_2 gaps and have significantly advanced the comprehensive understanding of the carbon cycle. However, these studies have mainly focused on specific regions such as China (C. He et al., 2022; Q. He et al., 2023; S. He et al., 2023; T. Li et al., 2023; W. Wang et al., 2022; C. Wu et al., 2023; M. Zhang & Liu, 2023; L. Zhang et al., 2022) and Iran (Siabi et al., 2019). Recently, J. Li et al. (2022) and L. Zhang, Li, et al. (2023) applied ML approaches to derive global spatial distributions of XCO_2 at 8-day and monthly temporal resolutions,

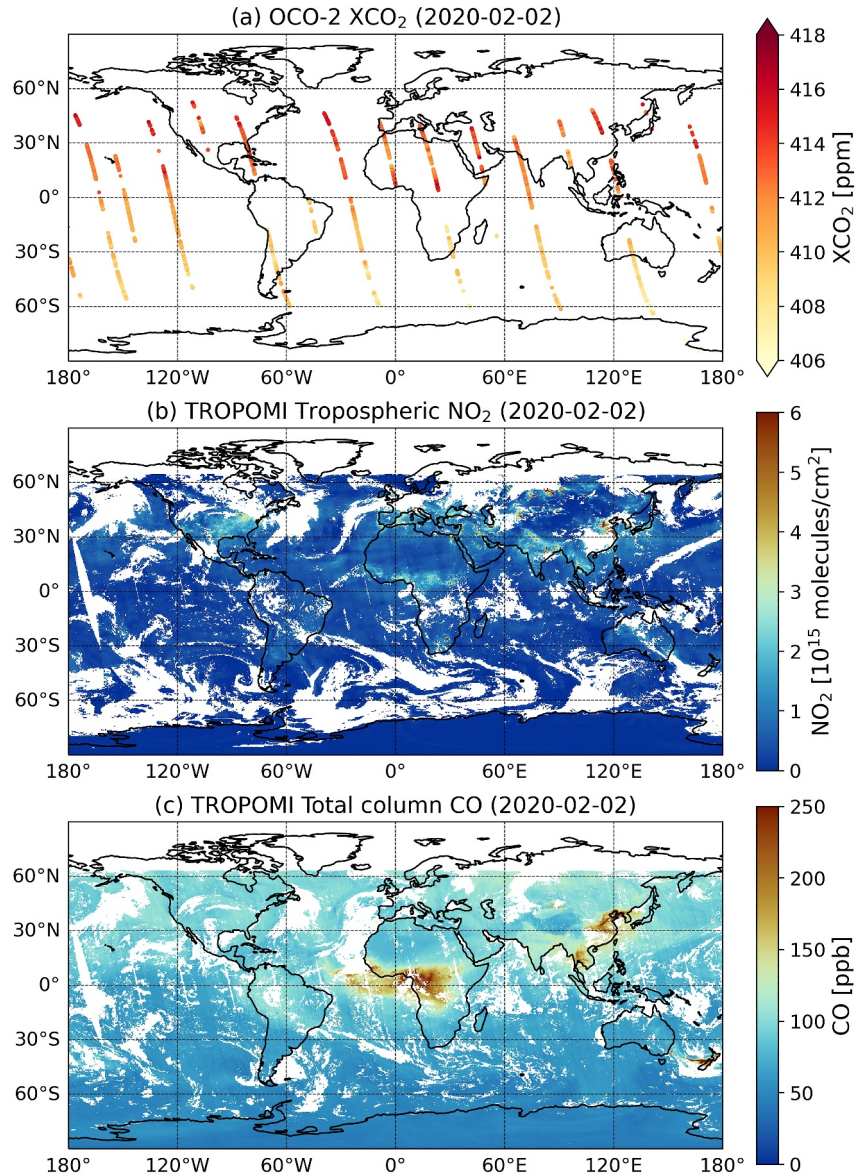


Figure 1. Spatial distribution of (a) OCO-2 XCO_2, (b) Sentinel-5P TROPOMI tropospheric NO_2 concentration, and (c) Sentinel-5P TROPOMI total column CO concentration measured on 2 February 2020.

respectively. However, these studies have been limited to retrieving XCO_2 data sets only over the land surface owing to constraints in the availability of training data sets, such as the Total Carbon Column Observing Network (TCCON) ground-based XCO_2 measurements. Moreover, the temporal resolutions of 8-day and 1-month in these studies may be inadequate for detecting fine-scale XCO_2 variations.

This study introduces an ML approach to estimate daily XCO_2 at a 0.25° spatial resolution globally using multiple-satellite measurements from the OCO-2, Sentinel-5 Precursor TROPOspheric Monitoring Instrument (Sentinel-5P TROPOMI), and ERA5 reanalysis data. Figure 1 illustrates the spatial distribution of the OCO-2 XCO_2, Sentinel-5P TROPOMI tropospheric nitrogen dioxide (NO_2) concentration, and TROPOMI total column carbon monoxide (CO) concentration, as measured on 2 February 2020. Although the OCO-2 provides highly accurate XCO_2 data on a global scale, its narrow swath results in many observation gaps (Figure 1a). In contrast, Sentinel-5P TROPOMI has a wide swath of approximately 2,600 km, enabling near-daily global coverage and more frequent observations (Figures 1b and 1c). Given that atmospheric CO_2 shares common combustion sources with NO_2 and CO (H. Park et al., 2021; Silva & Arellano, 2017; Sim et al., 2020; D. Wu

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et al., 2022; Yang et al., 2023), and considering near-daily global coverage from Sentinel-5P TROPOMI coupled with the high accuracy of OCO-2 XCO_2 data, this study employs the ML techniques to estimate daily XCO_2 on a global scale from May 2018 to December 2023. Considering the complex nonlinearity between XCO_2 and its influencing factors, we developed an ML model to estimate the daily XCO_2 on a global scale, hereafter referred to as ML XCO_2. This study fills the gap in satellite-based XCO_2 data on a global scale and introduces a novel method for producing daily XCO_2 estimates. This study provides invaluable insights for advancing our understanding of the carbon cycle.

The remainder of this paper is organized as follows. Section 2 provides a concise overview of the training and validation data. Section 3 describes the methodology used to train and validate the ML model. Section 4 details the results, including the ML model performance, spatiotemporal distributions of ML XCO_2, and the estimated atmospheric XCO_2 growth rates. Finally, Section 5 provides a comprehensive discussion and conclusion, summarizing the key insights and implications of this study.

2. Data

2.1. Training Data

OCO-2 provides highly accurate XCO_2 data with a precision of ±1 ppm, making it ideal for global scale studies (Eldering, O'Dell, et al., 2017). Additionally, OCO-2 collects approximately 100 times more samples daily than the GOSAT series, providing a much greater data density (Crisp et al., 2022; Eldering, O'Dell, et al., 2017). Given the high accuracy and data density of the OCO-2 XCO_2, these measurements were used as labels to estimate global daily XCO_2. For feature variables, we included the Sentinel-5P TROPOMI NO_2 tropospheric vertical column and CO total column products, which are commonly co-emitted with atmospheric CO_2. We also used the Sentinel-5P TROPOMI solar-induced fluorescence (SIF; Guanter, Bacour, Schneider, Aben, van Kempen, et al., 2021), which provides insights into vegetation activity. Additionally, ERA5 atmospheric variables, spatial variables (latitude and longitude), and temporal variables (year and day of the year (DOY)) were included as feature variables. Considering atmospheric CO_2 concentration relates to anthropogenic CO_2 emissions, it would be helpful to consider publicly available global greenhouse gas emission data set such as the Open-source Data Inventory for Anthropogenic CO_2 (ODIAC) (Oda et al., 2018; Oda & Maksyutov, 2011) or the Emissions Database for Global Atmospheric Research (EDGAR) (Crippa et al., 2024). For example, M. Zhang and Liu (2023) and C. Wu et al. (2024) used ODIAC CO_2 emissions to reconstruct monthly XCO_2 in China. However, ODIAC provides CO_2 emissions at a monthly temporal resolution, and at the time of this study, ODIAC data were only available until December 2019. Similarly, EDGAR primarily provides annual CO_2 emissions. Since our ML XCO_2 estimates cover the period from May 2018 to December 2023 on a daily timescale, the inclusion of ODIAC or EDGAR CO_2 emission data was less suitable for our purpose. Similarly, given that atmospheric CO_2 concentration is also influenced by vegetation activity, vegetation indices such as the Moderate Resolution Imaging Spectroradiometer (MODIS) Normalized Difference Vegetation Index (NDVI) or Enhanced Vegetation Index (EVI) could be considered. However, MODIS NDVI and EVI have a monthly temporal resolution, making them also less suitable for our purpose. Instead, we have incorporated Sentinel-5P daily TROPOMI SIF for vegetation activity.

The labels and feature variables used in this study are summarized in Table 1, and this section briefly describes the characteristics of these data sets.

2.1.1. OCO-2

OCO-2 (Crisp et al., 2017; Eldering, O'Dell, et al., 2017) is the first NASA's satellite dedicated to observing atmospheric CO_2 to enhance our understanding of the carbon cycle. Its grating spectrometers measure the high-resolution spectra of sunlight reflected from the Earth's surface across three spectral bands: 0.765 μm (O_2-A band), 1.61 μm (weak CO_2 band), and 2.06 μm (strong CO_2 band). These spectral measurements are processed using an optimal estimation-based algorithm (e.g., O'Dell et al., 2012) to retrieve the XCO_2. OCO-2 has a swath width of approximately 10 km and operates on a 16-day revisit cycle with a pixel size of 1.29×2.25 km^2 at nadir. In this study, we used level 2 bias-corrected XCO_2 full physics retrievals from the latest version (version 11), from May 2018 to December 2023, as labels for training our ML models. The XCO_2 data sets used in this study were retrieved using the Atmospheric Carbon Observations from Space (ACOS) algorithm (O'Dell et al., 2012). Here,

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Table 1
Summary of Variables Used for Training the Machine Learning Model in This Study

Type	Data set	Variables	Unit
Label	OCO-2	XCO_2	ppm
Features	Sentinel-5P TROPOMI	Nitrogen dioxide tropospheric column	mol/m^2
		Carbon monoxide total column	mol/m^2
		Solar-induced fluorescence (SIF) at 743–758 nm	$mW/m^2/sr/nm$
	ERA5	2 m temperature	K
		Sea surface temperature	K
		Surface pressure	Pa
		10 m u-component of wind	m/s
		10 m v-component of wind	m/s
		Latitude	degree
		Longitude	degree
		Year	-
		DOY (Day of the Year)	-

Note. Note that SIF was only used for land model training and sea surface temperature was only used for ocean model training.

we only used the high-quality XCO_2 measurements, indicated by an "xco2_quality_flag" of 0, to ensure the accuracy of our training data.

2.1.2. Sentinel-5P TROPOMI

The TROPOMI (Veefkind et al., 2012) onboard the Sentinel-5P satellite is a push-broom spectrometer designed to measure spectral data across multiple bands: ultraviolet-visible (270–500 nm), near-infrared (675–775 nm), and shortwave infrared (2,305–2,385 nm). This instrument retrieves the column abundances of atmospheric trace gases, such as NO_2, CO, CH_4, ozone, sulfur dioxide, and formaldehyde; cloud properties, including cloud cover, cloud optical thickness, and cloud top height; and aerosol properties, such as aerosol index and aerosol layer height. TROPOMI has a swath width of approximately 2,600 km, providing nearly daily global coverage. The nadir pixel size was improved from 3.5×7.0 km^2 to 3.5×5.5 km^2 on 6 August 2019.

In this study, we used the TROPOMI level 2 NO_2 tropospheric vertical column, CO total column, and SIF products from May 2018 to December 2023 to train the ML models. To ensure high-quality measurements in our analysis, we applied a "qa_value" threshold of 0.75 for the TROPOMI NO_2 tropospheric vertical column product and 0.5 for the TROPOMI CO total column and SIF products, following the recommendations in the Sentinel-5 precursor/TROPOMI Product User Manual (Apituley et al., 2022; Eskes et al., 2022; Guanter et al., 2022).

2.1.3. ERA5

ERA5, developed by the European Centre for Medium-Range Weather Forecasts (ECMWF), represents the latest climate reanalysis data and is essential for global climate research and meteorological applications (Hersbach et al., 2020). This data set was produced using data assimilation techniques that combine newly available observations with previous model forecasts to generate the most probable state of the atmosphere. From 1940 to the present, ERA5 has provided hourly estimates for a large number of atmospheric, oceanic, and land-surface quantities, offering a comprehensive and consistent global atmospheric data set. The ERA5 data set has a native spatial resolution of approximately 31 km and is available on a regular latitude-longitude grid of 0.25° from the Copernicus Climate Data Store (https://cds.climate.copernicus.eu/).

In our study, the hourly ERA5 data sets were aggregated into daily time steps from May 2018 to December 2023. The variables used were the 2 m temperature, sea surface temperature, surface pressure, 10 m u-component of wind, and 10 m v-component of wind as detailed in Table 1.

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Table 2
List of TCCON Sites Used for Ground-Based Validation of ML XCO_2

Station	Latitude	Longitude	Start date	End date
Bremen	53.104°N	8.850°E	6 January 2009	24 June 2021
Burgos	18.553°N	120.650°E	3 March 2017	23 June 2023
Caltech	34.140°N	118.130°W	20 September 2012	25 February 2024
Darwin	12.456°S	130.927°E	1 January 2013	27 December 2022
East Trout Lake	54.354°N	104.987°W	3 October 2016	2 June 2024
Edwards	34.95°N	117.88°W	20 July 2013	22 February 2024
Garmisch	47.476°N	11.063°E	18 July 2007	4 May 2023
Hefei	31.90°N	117.17°E	2 November 2015	25 December 2023
Harwell	51.57°N	1.32°W	30 May 2021	31 March 2024
Izaña	28.3°N	16.5°W	2 January 2014	30 August 2023
Karlsruhe	49.100°N	8.439°E	15 January 2014	26 June 2023
Lauder	45.038°N	169.684°E	2 October 2018	28 December 2023
Lamont	36.604°N	97.486°W	16 April 2011	25 February 2024
Nicosia	35.14°N	33.38°E	1 September 2019	10 May 2023
Orléans	47.997°N	2.113°E	6 September 2019	23 June 2023
Paris	48.850°N	2.356°E	23 September 2014	20 December 2023
Park Falls	45.945°N	90.273°W	2 June 2004	25 February 2024
Rikubetsu	43.457°N	143.766°E	24 June 2014	27 June 2023
Saga	33.241°N	130.288°E	28 July 2011	14 October 2022
Tsukuba	36.051°N	140.122°E	28 March 2014	31 March 2021
Wollongong	34.406°S	150.879°E	4 January 2013	27 June 2023
Xianghe	39.800°N	116.960°E	14 June 2018	29 May 2023

Note. Note that start and end date refer to data range where TCCON GGG2020 retrievals are available.

2.2. Validation Data: TCCON

To ensure the accuracy of ML XCO_2 retrievals, thorough validation is essential. In this study, ML XCO_2 retrievals were validated using 10-fold cross-validation and ground-based TCCON data sets (Wunch et al., 2011). Details on 10-fold cross-validation are provided in Section 3.3.

TCCON is a prominent global observation network of ground-based solar-viewing FTSs that records direct solar absorption spectra in the near-infrared spectral region. TCCON sites are distributed globally, including in North America, Europe, Asia, and Oceania. The high spectral resolution of the FTSs used in TCCON enables precise and accurate measurements of atmospheric trace gases, contributing to a comprehensive understanding of their distribution and behavior in the atmosphere and playing a crucial role in validating satellite data (X. Hong et al., 2022; Wunch et al., 2017; Yoshida et al., 2013). The TCCON near-infrared spectral data were processed using the GFIT nonlinear least-squares fitting algorithm, which is the core component of the GGG processing software to retrieve the column-averaged dry air mole fractions of atmospheric constituents such as CO_2, CH_4, and CO.

Recently, the TCCON data processed with the latest GGG2020 software were publicly released in April 2022 (Laughner et al., 2023). In this study, we used this latest TCCON station measurements (GGG2020) that available since May 2018 to validate ML XCO_2 retrievals. Table 2 summarizes the specific information for the TCCON sites used in this study.

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3. Methods

3.1. Data Processing

As mentioned in Section 2.1, we used OCO-2 XCO_2 as the label, and the TROPOMI NO_2 tropospheric vertical column, CO total column, and ERA5 atmospheric variables as feature variables. However, these variables had different spatial and temporal resolutions. Therefore, matching the resolution of the data was necessary to prepare the data set. In this study, all data sets used to train the ML model were processed to a regular latitude-longitude grid of 0.25°. This resolution was selected to align with the spatial resolution of the ERA5 reanalysis data. For temporal resolution, the hourly ERA5 data sets were aggregated into daily time steps.

There are many sources of uncertainty in high-latitude OCO-2 retrieval. For instance, high solar zenith angles affect the slant column path of radiation. In addition, over surface covered with snow and ice at high latitudes, OCO-2 retrievals often fail because of high surface reflectance (Eldering, O'Dell, et al., 2017; Jacobs et al., 2024). Consequently, this study restricted the model training to data from latitudes between 60°S and 60°N to avoid uncertainty in these high-latitude regions.

3.2. Training of ML Model

Based on the previously mentioned label and feature variables (Table 1), we trained a Random Forest (RF) model (Breiman, 2001) to estimate the global daily XCO_2 from May 2018 to December 2023. The RF is an ensemble algorithm that constructs multiple decision trees from bootstrap sampling and outputs the average prediction of individual trees. Owing to its robustness and accuracy, RF is particularly effective in managing large data sets with high dimensionality. Additionally, it mitigates the issue of overfitting, which is common in single-decision tree models by averaging the predictions across various trees. In addition to the RF model, we trained various deep-learning-based models, including an ANN and one-dimensional Convolution Neural Network (CNN). ANN and CNN models are also widely used in various environmental modeling processes (Han & Wang, 2021; Kareem et al., 2021; Tahmasebi et al., 2020). Detailed descriptions of training ANN and CNN models are provided in Text S1 in Supporting Information S1. During the training of all three models, we performed hyperparameter tuning, and the results are summarized in Table S1 in Supporting Information S1.

3.3. Model Validation

As mentioned in Section 2.2, the ML XCO_2 retrievals were validated from two perspectives: 10-fold cross-validation and ground-based validation using TCCON data sets. In the 10-fold cross-validation approach, the data set was randomly divided into 10 separate folds of equal size. Each fold was used as a test data set and the remaining nine folds served as the training data set. The training data set was used for model training and the performance of the trained model was evaluated using a test data set. This process was repeated to ensure that each fold was used exactly once as the test data set, allowing for a fair evaluation of the model performance across all data. For ground-based validation, we utilized TCCON XCO_2 measurements, which are highly accurate and commonly used to validate satellite-derived XCO_2 data such as OCO-2 and GOSAT (Wunch et al., 2011). We only included TCCON sites situated between 60°S and 60°N that have been providing data since May 2018, the start of the ML XCO_2 data series.

To quantitatively assess the ML model, we used three statistical metrics: correlation of determination (R^2), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE), which are defined as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{n} \left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{n} \left(y_i - \bar{y}\right)^2} \tag{1}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2} \tag{2}$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \hat{y}_i\right| \tag{3}$$

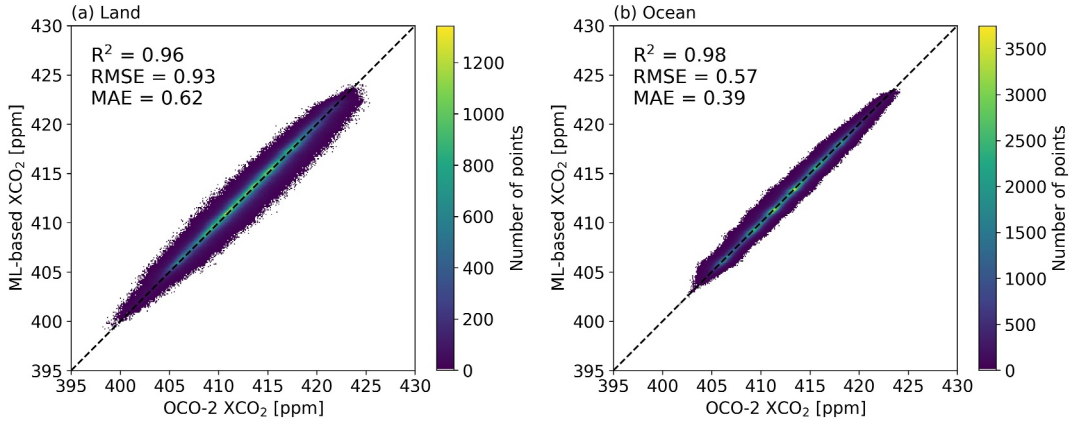


Figure 2. Scatter density plots from 10-fold cross-validation results showing the relationship between OCO-2 XCO_2 and ML XCO_2 estimated from the Random Forest model over (a) land and (b) ocean. The dashed lines represent the 1:1 line.

where y_i and $\hat{y}_i$ represent the observed and estimated XCO_2 values from OCO-2 and the ML model for sample i, respectively. $\bar{y}$ is the mean observed XCO_2. n is the number of data sets.

3.4. Estimation of ML XCO_2

After training the ML model as described in Section 3.3, daily inputs such as the Sentinel-5P TROPOMI NO_2 tropospheric vertical column, CO total column, SIF products, and ERA5 reanalysis data (feature variables in Table 1) were fed into the trained ML model to estimate the ML XCO_2. An example of the ML XCO_2 estimation on a global scale for 1 January 2023 is shown in Figure S1 in Supporting Information S1. The daily TROPOMI NO_2 tropospheric vertical column and CO total column concentrations (Figures S1a and S1b in Supporting Information S1), along with the daily ERA5 reanalysis data (not shown), as described in Table 1, were utilized as inputs to the trained ML model to estimate ML XCO_2 (Figure S1c in Supporting Information S1).

4. Results

4.1. Model Performance

We first evaluated the performance of the model in terms of 10-fold cross-validation and ground-based TCCON measurements as described in Section 3.3. Figure 2 presents the scatter density plots from the 10-fold cross-validation, illustrating the relationship between the OCO-2 XCO_2 measurements and ML XCO_2 estimations from the RF model over land and ocean. The validation metrics were a R^2 of 0.96, a RMSE of 0.93 ppm, and a MAE of 0.62 ppm over land. Over the ocean, the model achieved a R^2 of 0.98, a RMSE of 0.57 ppm, and a MAE of 0.39 ppm. These results demonstrate the robustness of the model for reconstructing daily XCO_2 on a global scale.

As mentioned in Section 3.2, we also evaluated the performances of the ANN and CNN models using a 10-fold cross-validation (Figures S2 and S3 in Supporting Information S1). In our evaluations, the RF, ANN, and CNN models demonstrated robust performances; however, the RF model performed slightly better in terms of 10-fold cross-validation (Table S2 in Supporting Information S1). Consequently, the results presented henceforth are based on the application of the RF model.

Figure 3 displays scatterplots that compare the ML daily averaged XCO_2 estimations to those from the TCCON ground-based measurements across 22 different stations (Table 2). Each color in Figure 3 corresponds to the result from a different TCCON station, illustrating a comprehensive validation effort. Figure 3 illustrates the high degree of agreement across the diverse distributions of TCCON stations, confirming the accuracy of the ML model in estimating XCO_2 on a global scale. The overall validation metrics were an R^2 of 0.95, RMSE of 1.05 ppm, and MAE of 0.78 ppm. Table 3 provides a detailed comparison between ML XCO_2 and TCCON XCO_2 for each station, further highlighting the model's consistently high accuracy across various geographic and atmospheric conditions and its reliability for global applications. It is noted that the accuracy of ML XCO_2 at the Xianghe site appears relatively lower than that at other sites (R^2 = 0.85, RMSE = 1.79 ppm, MAE = 1.42 ppm, as

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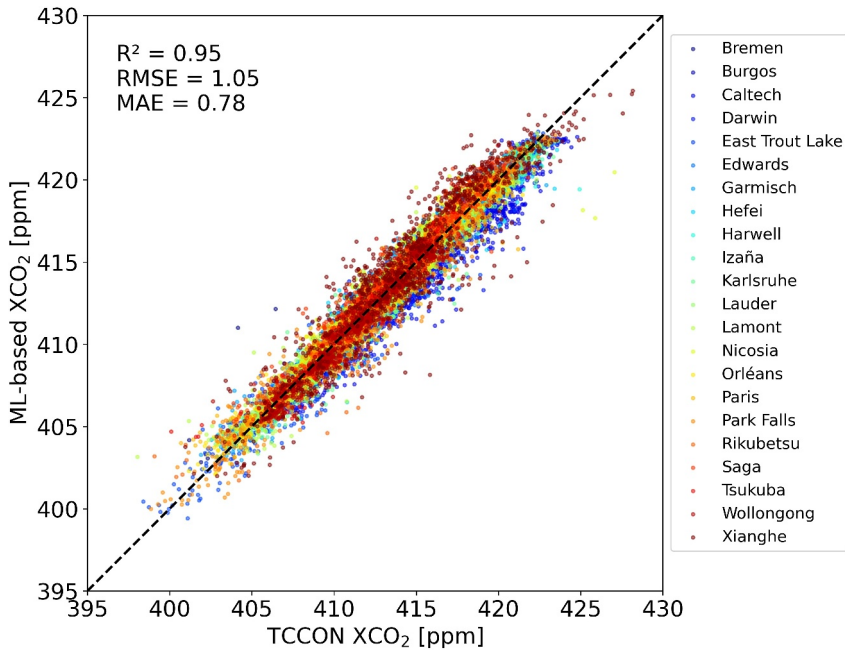


Figure 3. Scatterplot comparing TCCON XCO_2 measurements against ML XCO_2 estimations from the Random Forest model across various TCCON sites. Each color represents results from a different TCCON station as indicated in the legend. The dashed lines represent the 1:1 line.

shown in Table 3). These statistical metrics are comparable to those reported in previous studies (Hua et al., 2024; W. Liu et al., 2024; Y. Wang et al., 2023; C. Wu et al., 2024; L. Zhang, Li, & Wu et al., 2022). One possible explanation is aerosol interference, which is known to be one of uncertainty sources in OCO-2 XCO_2 retrievals (Connor et al., 2016; Virtanen et al., 2025; Wunch et al., 2017). Scattering by aerosols affects the light path of radiation, thereby impacting the retrieved XCO_2 from satellite observations. Since the Xianghe site is located near Beijing, China, which suffers from severe air pollution, aerosol contamination may still affect OCO-2 XCO_2 retrievals at this site. Although we used quality-filtered OCO-2 XCO_2 data as labels in ML XCO_2, some residual uncertainties caused by aerosols may remain.

Figure 4 presents examples of the daily averaged time series from ML XCO_2 (red lines) and TCCON XCO_2 (blue lines) at four selected TCCON sites across different continents: Park Falls (North America), Karlsruhe (Europe), Saga (Asia), and Wollongong (Oceania). In Park Falls, Karlsruhe, and Saga, located in the Northern Hemisphere (Table 2), both TCCON and ML XCO_2 exhibited strong seasonal CO_2 fluctuations. These fluctuations are primarily influenced by anthropogenic CO_2 emissions and CO_2 uptake by the terrestrial biosphere. Meanwhile, in Wollongong TCCON site, located in the Southern Hemisphere (Table 2), both TCCON and ML XCO_2 showed weaker seasonal CO_2 fluctuations, owing to the region's limited anthropogenic CO_2 emissions and less seasonal variation in vegetation activity (Eldering, Wennberg, et al., 2017). Figure 4 demonstrates that the temporal ML XCO_2 variations were consistent with those from the TCCON measurements. Detailed XCO_2 time-series comparisons for all the TCCON stations are provided in Figure S4 in Supporting Information S1, further confirming the overall agreement between the ML XCO_2 and TCCON XCO_2 time-series data.

4.2. Spatiotemporal Distribution of ML XCO_2

Based on the validated high accuracy of the ML model for reconstructing daily XCO_2 on a global scale, as detailed in Section 4.1, we investigated the spatiotemporal distribution of the ML XCO_2. The multiyear-averaged global ML XCO_2 spatial distributions for the complete years from 2019 to 2023 are shown in Figure S5 in Supporting Information S1. This multiyear averaged ML XCO_2 map clearly shows that XCO_2 levels in the Northern Hemisphere are generally higher than those in the Southern Hemisphere. Specifically, regions, such as China, the United States, and India, which have high anthropogenic CO_2 emissions (Friedlingstein et al., 2023), exhibit relatively high XCO_2 levels. Additionally, regions such as Northern South America and Central Africa also

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Table 3
Statistics of Ground-Based Validation With TCCON Stations, Detailing the Performance Metrics Including R^2, RMSE, and MAE for Each Site

Station	R^2	RMSE (ppm)	MAE (ppm)
Bremen	0.93	1.20	0.83
Burgos	0.95	0.76	0.62
Caltech	0.90	1.39	1.10
Darwin	0.83	1.21	1.06
East Trout Lake	0.96	1.10	0.83
Edwards	0.98	0.61	0.46
Garmisch	0.94	1.07	0.87
Hefei	0.92	1.12	0.87
Harwell	0.89	0.93	0.65
Izaña	0.98	0.64	0.51
Karlsruhe	0.96	1.03	0.77
Lauder	0.95	0.64	0.51
Lamont	0.96	0.91	0.69
Nicosia	0.93	0.93	0.66
Orléans	0.97	0.86	0.66
Paris	0.96	0.97	0.70
Park Falls	0.96	1.03	0.77
Rikubetsu	0.93	1.15	0.82
Saga	0.92	1.07	0.86
Tsukuba	0.92	1.01	0.79
Wollongong	0.94	0.75	0.55
Xianghe	0.85	1.79	1.42

showed relatively high XCO_2 levels, which were likely affected by natural emissions such as wildfires and biomass burning. These spatial patterns of the multiyear averaged global ML XCO_2 are consistent with findings from previous studies (Eldering, Wennberg, et al., 2017; Sheng et al., 2023; Y. Wang et al., 2023).

Figure 5 shows the seasonally averaged global ML XCO_2 spatial distributions from 2019 to 2023 categorized by the following periods: March-April-May (MAM), June-July-August (JJA), September-October-November (SON), and December-January-February (DJF). During DJF, which corresponds to the winter months in the Northern Hemisphere, regions with high anthropogenic emissions, such as Eastern China, exhibited relatively high XCO_2 levels. This period was characterized by minimal CO_2 uptake and increased CO_2 emissions from burning fossil fuels, contributing to elevated XCO_2. XCO_2 levels in the Northern Hemisphere reached their seasonal peaks during MAM corresponding to the spring months. This peak occurred because the temperatures had not yet increased sufficiently to reawaken the biosphere owing to low activity of winter (Eldering, Wennberg, et al., 2017). During JJA and SON, which correspond to the summer and autumn months in the Northern Hemisphere, respectively, the XCO_2 levels are relatively lower in the Northern Hemisphere because of the significant uptake of CO_2 by the terrestrial biosphere, although anthropogenic emissions continue. A similar but opposite pattern of XCO_2 variation was observed in the Southern Hemisphere. However, these seasonal variations in XCO_2 are more evident in the Northern Hemisphere primarily because of extensive land masses and significant plant growth cycles influencing CO_2 fluctuations (Eldering, Wennberg, et al., 2017). This seasonal disparity between the hemispheres is further supported by the findings shown in Figure 4. The patterns of seasonal variation in global ML XCO_2 are consistent with those observed by OCO-2 satellite (Eldering, Wennberg, et al., 2017).

As mentioned in Section 3.1, the ML model was trained using data sets between 60°S and 60°N because satellite measurements tend to have larger uncertainties at higher latitudes. ML XCO_2 for latitudes beyond ±60° were estimated based on the relationships trained from data sets between 60°S and 60°N. It indicates that ML XCO_2 estimates at high-latitude regions may be less robust. Therefore, we have excluded ML XCO_2 validation and their quantitative analyses at high latitude regions from our validation analysis.

Figure 6 displays the global monthly averaged time series over land for the complete years from 2015 to 2023 comparing ML XCO_2 (blue line) with operating satellite retrievals (OCO-2 XCO_2 in green, OCO-3 XCO_2 in pink, and GOSAT XCO_2 in red) and the model-based products including Copernicus Atmospheric Monitoring Service (CAMS) XCO_2 reanalysis data (CAMS-EGG4; black line), CarbonTracker (CT2022; Jacobson et al., 2023) (brown line), and CarbonTracker Near-Real Time (CT-NRT.v2024-1; Jacobson et al., 2024) (cyan line) between 60°S and 60°N. The ML XCO_2 global monthly averaged timeseries exhibited an upward trend and strong seasonal CO_2 fluctuations with the monthly average increasing from 406.48 ppm in May 2018 to 420.77 ppm in December 2023. The ML XCO_2 global monthly averaged time series is consistent with that of other established XCO_2 products. Note that in 2020, ML XCO_2, as well as satellite-based XCO_2 products (OCO-2, OCO-3 and GOSAT XCO_2) and CT2022/CT-NRT.v2024-1 products were underestimated compared with the CAMS reanalysis XCO_2. This discrepancy can be explained by the fact that the anthropogenic CO_2 emissions used to estimate XCO_2 in the CAMS reanalysis were not adjusted for the Coronavirus disease 2019 (COVID-19) related CO_2 emission reductions in 2020 (Le Quéré et al., 2020) resulting in an overestimation of CAMS XCO_2 for that year. Future updates by CAMS research teams are expected to address this issue (Agustí-Panareda et al., 2023).

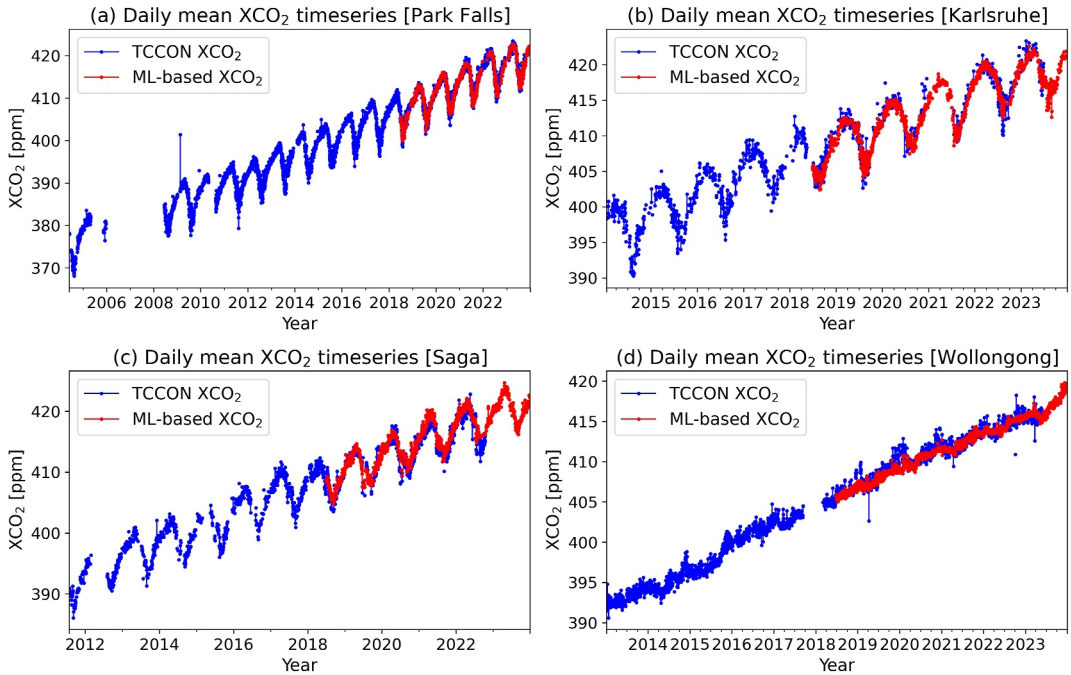


Figure 4. Daily averaged XCO_2 timeseries at four selected TCCON sites across different continents: (a) Park Falls (North America), (b) Karlsruhe (Europe), (c) Saga (Asia), and (d) Wollongong (Oceania). The blue lines indicate TCCON daily averaged XCO_2 timeseries, whereas the red lines represent ML XCO_2 daily averaged timeseries.

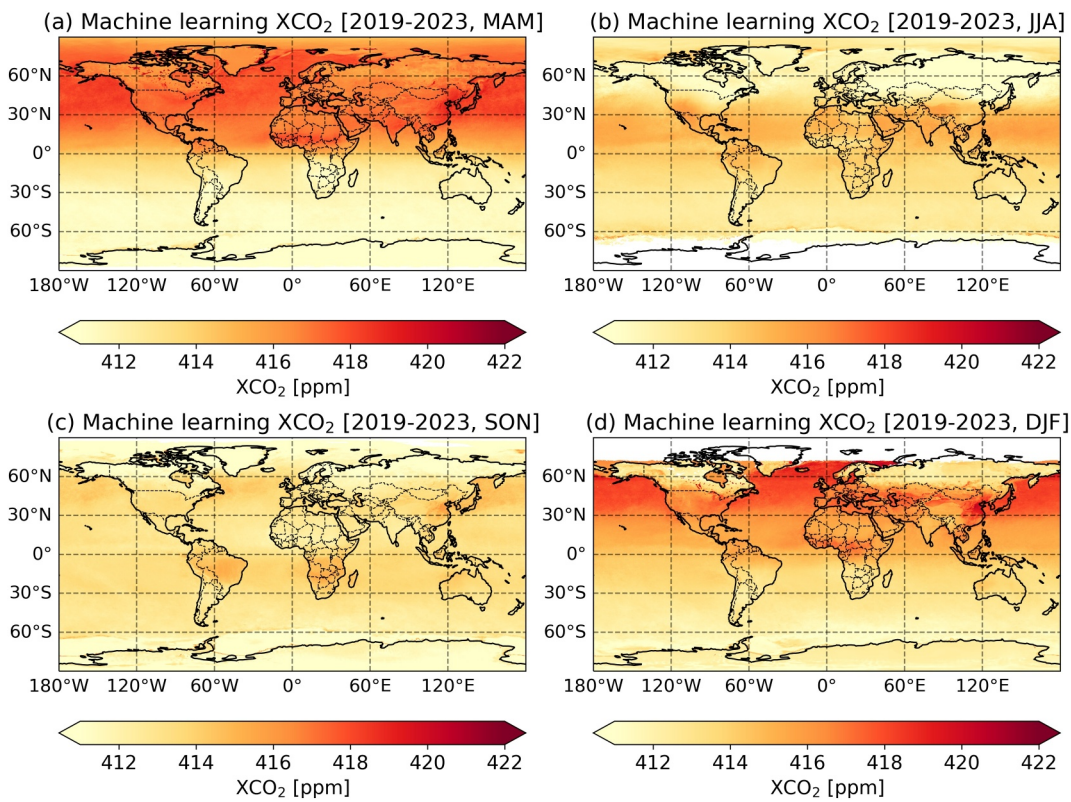


Figure 5. Seasonal averaged ML XCO_2 spatial distributions for the complete years from 2019 to 2023: (a) March-April-May (MAM), (b) June-July-August (JJA), (c) September-October-November (SON), and (d) December-January-February (DJF).

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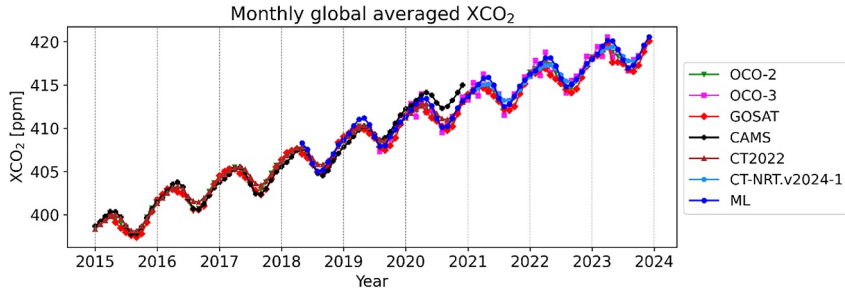


Figure 6. Monthly global averaged XCO_2 monthly timeseries over land from 2015 to 2023 between 60°S and 60°N. The green, pink, and red lines indicate satellite-based XCO_2 timeseries from OCO-2, OCO-3, and GOSAT, respectively. The black, brown, and cyan lines represent model-based XCO_2 timeseries from CAMS, CarbonTracker (CT2022), and CarbonTracker Near-Real Time product (CT-NRT.v2024-1), respectively. The blue line depicts the ML XCO_2 timeseries.

4.3. Spatial Comparisons of ML XCO_2 With CarbonTracker and CAMS Products

We validated ML XCO_2 with ground-based TCCON XCO_2 (Section 4.1). To further investigate the spatial reliability of ML XCO_2, we compared the ML XCO_2 with model-based products including CT2022 and CAMS XCO_2 from 2019 to 2020 during which all data sets are available (Figure S6 in Supporting Information S1). The general spatial patterns of CT2022, CAMS, and ML XCO_2 are consistent on a global scale, although regional differences are shown in certain areas such as Eastern China. In 2020, CAMS XCO_2 (Figure S6b in Supporting Information S1) were generally higher than CT2022 and ML XCO_2 (Figures S6d and S6f in Supporting Information S1), which is consistent with Figure 6. Figure 7 shows an example of regional comparison of ML XCO_2, CAMS XCO_2, and CT2022 XCO_2 from 2019 to 2020 over East Asia. Compared to CAMS XCO_2 (0.75° spatial resolution) and CT2022 XCO_2 (3° × 2° spatial resolution), ML XCO_2 offers a higher spatial resolution (0.25° spatial resolution), therefore, more effectively provides a finer-scale representation of XCO_2 on a regional scale.

Figures S7 to S9 in Supporting Information S1 show the scatter density plots comparing ML XCO_2 with CT2022, CAMS XCO_2 as well as OCO-2/3 XCO_2 from 2019 to 2020 during which all data sets are available. For a fair comparison, we only considered grid points where all data sets are available. We conducted the analysis separately for three cases: (a) over land and ocean (Figure S7 in Supporting Information S1), (b) over land (Figure S8 in Supporting Information S1), and (c) over ocean (Figure S9 in Supporting Information S1). Our analysis indicates that ML XCO_2 shows the highest agreement with OCO-2 XCO_2, because it was trained using OCO-2 XCO_2 as labels. Additionally, ML XCO_2 exhibits high consistencies with OCO-3 and CarbonTracker (CT2022) XCO_2. In contrast, CAMS XCO_2 tends to be generally overestimated compared to ML XCO_2. One possible explanation is that CAMS reanalysis was not adjusted for COVID-19 related CO_2 emission reductions in 2020 (Agustí-Panareda et al., 2023) as explained in Figure 6. In addition, changes from reprocessed satellite products to the near-real-time products in 2019 affected the quality of the observations assimilated in CAMS

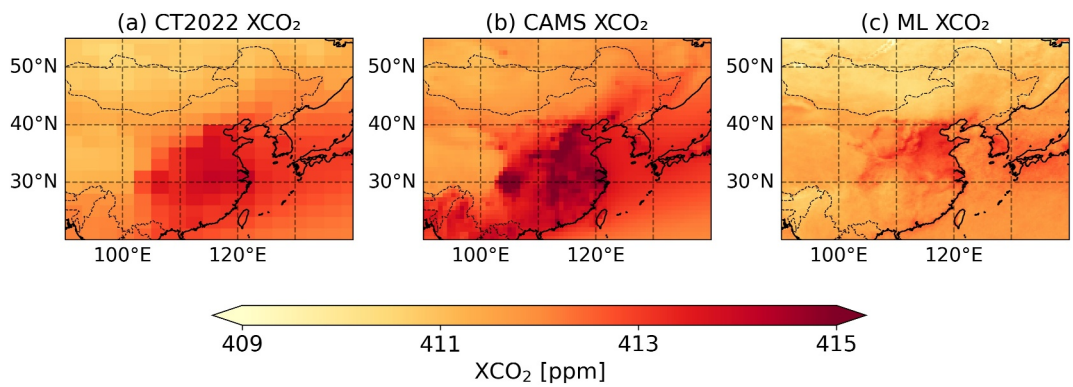


Figure 7. Spatial distributions of (a) CarbonTracker (CT2022), (b) CAMS, and (c) ML XCO_2 from 2019 to 2020 over East Asia.

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reanalysis, thereby impacting the quality of CAMS greenhouse gases reanalysis from 2019 (Agustí-Panareda et al., 2023).

4.4. Annual Mean Growth Rate of XCO_2

We estimated the annual mean growth rate of XCO_2 using the ML XCO_2 data sets from 2019 to 2023. This calculation follows the method suggested by Buchwitz et al. (2018), which is similar to the approach of Thoning et al. (1989) used by the National Oceanic and Atmospheric Administration (NOAA) Global Monitoring Laboratory (GML) for surface CO_2 measurements but specifically designed for satellite-based XCO_2 data sets. In this method, the monthly XCO_2 time series is first computed. The annual growth rate for each month was calculated as the difference between the XCO_2 value of that month and that of the same month of the previous year with the result assigned to the midpoint of the corresponding 1-year period. The annual mean growth rate of XCO_2 was determined by averaging the monthly growth rates. For more detailed information, readers are referred to Buchwitz et al. (2018).

Figure 8a–8e present the estimated spatial distributions of the annual mean XCO_2 growth rate over land from 2019 to 2023 between 60°S and 60°N (left panels), with the corresponding zonal mean growth rates (right panels) for each figure. The spatial distributions of the annual mean XCO_2 growth rate show the variation in the annual mean XCO_2 growth rate across different regions, highlighting areas with both higher and lower growth rates. The estimated global average annual mean XCO_2 growth rates vary annually with values consistently above 2.0 ppm/year. The global average annual mean XCO_2 growth rates were 2.735, 2.423, 2.389, 2.065, and 2.487 ppm/year in 2019, 2020, 2021, 2022, and 2023, respectively. These growth rates and their variation patterns from 2019 to 2023 are in good agreement with those estimated using the OCO-2 satellite data and NOAA GML surface measurements (Figure S10 in Supporting Information S1). This result demonstrates the reliability and effectiveness of our ML approach for capturing trends in atmospheric CO_2.

In the regional analysis, an overall decrease in growth rates was observed in many regions in 2020 compared with 2019. However, despite the reduction of the anthropogenic CO_2 emissions in 2020 due to the global COVID-19 lockdowns, the change in XCO_2 growth rate (a reduction of approximately 0.3 ppm/year on a global scale) was within the natural interannual variability (Figure S10 in Supporting Information S1). This result indicates that there was no clear decline in the XCO_2 growth rate owing to the COVID-19-related decrease in CO_2 emissions. This result is consistent with the findings of previous studies (Laughner et al., 2021; Y. Liu et al., 2023). In 2021, XCO_2 growth rates generally remained at levels similar to those in 2020. In contrast, in 2022, the XCO_2 growth rates exhibited an overall reduction across many regions and showed the lowest growth rates from 2019 to 2023. This decrease in growth rates occurred even though global CO_2 emissions increased by 1.5% in 2022 relative to 2021 (Z. Liu et al., 2023). Although a detailed analysis of this phenomenon is beyond the scope of our study, the relatively lower XCO_2 growth rate in 2022 may be explained by the increased atmospheric CO_2 uptake due to the La Niña event during that year, which will be further discussed in Section 5.

We additionally estimated the annual mean growth rates of XCO_2 using CarbonTracker products and compared those estimated from ML XCO_2 from 2019 to 2023. Ideally, a single CarbonTracker product would be used for the entire analysis period. However, due to data availability constraints, we used CT2022 for the years 2019–2020 and CT-NRT.v2024-1 for 2021–2023 to compare with ML XCO_2 (Figures 8f–8j). The spatial distribution of the annual mean growth rates varies between both data sets. One possible explanation is the different spatial resolutions of the data sets, as CarbonTracker product has a coarser resolution ($3° \times 2°$ spatial resolution) compared to ML XCO_2 (0.25° spatial resolution). Despite these differences, their zonal mean growth rates are generally consistent with each other. Notably, the decline in the XCO_2 growth rate in 2022 and its increase again in 2023 are evident in both data sets.

5. Discussion and Conclusion

In this study, we propose an ML approach to estimate daily 0.25° XCO_2 data on a global scale, utilizing multiple-satellite measurements from OCO-2 and Sentinel-5P TROPOMI with ERA5 reanalysis data (Table 1) for the period from May 2018 to December 2023. We employed various models for this purpose, including the RF model and deep learning models such as the ANN and CNN models.

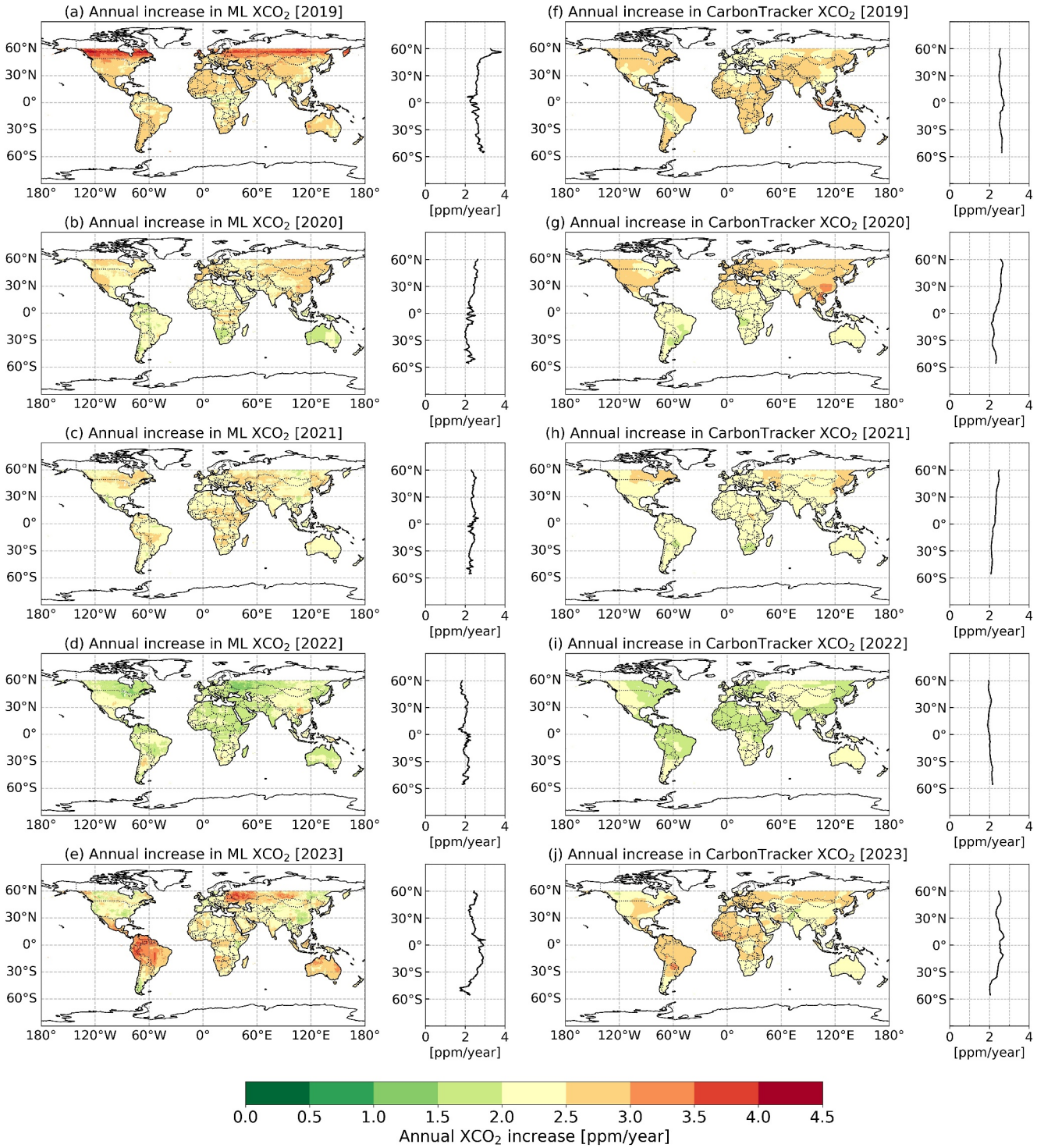


Figure 8. Annual mean atmospheric XCO_2 growth rates (unit: ppm/year) over land from 2019 to 2023 between 60°S and 60°N. Panels (a–e) present (left) spatial distributions and (right) corresponding zonal mean growth rates from ML XCO_2, whereas panels (f–j) show the corresponding results from CarbonTracker XCO_2.

Our validation began with 10-fold cross validation, which is a widely used method for evaluating the generalization capability of ML models. The cross-validation results indicate that the RF model achieved excellent performance metrics (Figure 2). These results underscore the robustness of the RF model for reconstructing daily XCO_2 on a global scale. We also conducted a ground-based validation using TCCON measurements, which are

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considered highly accurate for verifying satellite-based XCO_2 data. This validation demonstrated a high level of agreement between the ML XCO_2 estimations and XCO_2 measurements across multiple TCCON sites (Figures 3 and 4, Table 3, and Figure S4 in Supporting Information S1). This validation further confirmed the accuracy of ML XCO_2 estimations compared with reliable ground-based measurements across various geographic and atmospheric conditions.

The ML XCO_2 revealed distinct spatial and seasonal variation patterns from 2019 to 2023, as illustrated in Figure 5 and Figure S5 in Supporting Information S1, which were consistent with those reported in previous studies. In addition, the temporal variations in ML XCO_2 are consistent with existing XCO_2 products from OCO-2, OCO-3, GOSAT, CAMS, CarbonTracker, and CarbonTracker-Near-Real Time (Figure 6). The consistency in both the spatial and temporal variation patterns underscores the reliability of ML XCO_2 in accurately capturing global CO_2 dynamics.

Finally, we estimate the annual mean growth rates from the ML XCO_2 data set from 2019 to 2023 (Figure 8). During this period, the estimated annual mean growth rates ranged from 2.065 to 2.735 ppm/year, which were consistent with those from both OCO-2 satellite-based XCO_2 and NOAA GML surface-based CO_2 measurements (Figure S10 in Supporting Information S1). Previous studies (e.g., Buchwitz et al., 2018; Kim et al., 2016; Weir et al., 2021) have noted that changes in the atmospheric CO_2 growth rate are mainly driven by the El Niño Southern Oscillation (ENSO) cycle. During El Niño events, tropical land regions generally become warmer and dryer leading to a decrease in atmospheric CO_2 uptake. This resulted in higher CO_2 growth rates. During the strong El Niño event in 2015–2016, higher CO_2 growth rates were observed in both satellite- and ground-based CO_2 measurements (Figure S10 in Supporting Information S1), which is consistent with previous studies (e.g., Buchwitz et al., 2018). Conversely, La Niña events have the opposite effect resulting in lower CO_2 growth rates. The trend of a relatively low XCO_2 growth rate in 2022 (Figure 8d) was also confirmed by NOAA GML surface CO_2 measurements (Figure S10 in Supporting Information S1), which may be attributed to increased atmospheric CO_2 uptake during the 2022 La Niña event. Although the ML approach used in this study enables the estimation of annual mean XCO_2 growth rate distributions, a detailed analysis of these phenomena is beyond the scope of this study. Further research is required to fully understand the mechanisms driving observed growth rate variations.

The effectiveness of the ML approach in filling the gaps in the global XCO_2 data is demonstrated in Figure S11 in Supporting Information S1 and Figure 9. An example of the spatial distributions of OCO-2 XCO_2 and ML XCO_2 on 1 January 2023 is illustrated in Figure S11 in Supporting Information S1. As previously mentioned, although the OCO-2 satellite provides high-quality XCO_2 retrievals, it exhibits many observation gaps (Figure S11a in Supporting Information S1) because of its narrow swath width of approximately 10 km and disruptions caused by clouds and high aerosol levels. In contrast, ML XCO_2 provides broader area coverage for XCO_2 data (Figure S11b in Supporting Information S1), effectively filling the observation gaps using multiple satellite measurements from OCO-2 and Sentinel-5P TROPOMI, thereby overcoming the limitations of currently operating CO_2 monitoring satellites. Figure 9 presents the spatial distributions and probability density functions (PDFs) of the yearly averaged OCO-2 and ML XCO_2 over the United States, China, and Europe for 2023. Panels (a), (d), and (g) show the annually averaged spatial distributions of OCO-2 XCO_2, exhibiting observation gaps even when the XCO_2 data were aggregated over an entire year. In contrast, the ML XCO_2 spatial distributions (panels (b), (e), and (h)) demonstrate the continuous coverage of XCO_2 for the same regions filling the observation gaps. When comparing the annual spatial distributions of OCO-2 and ML XCO_2 with their corresponding PDFs (panels (c), (f), and (i)), some differences were noted. For OCO-2 XCO_2, the spatial distributions of the annual averaged XCO_2 were more discontinuous than those of ML XCO_2, with relatively larger differences between neighboring areas. This discrepancy is also evident in the corresponding PDFs. Such differences may be attributed to the sampling limitations of OCO-2 XCO_2 with many observation gaps and a low temporal resolution of 16 days, resulting in an incomplete representation of the annual spatial characteristics of XCO_2. In contrast, ML XCO_2, by filling these gaps, provides a more continuous and detailed description of the XCO_2 spatial distribution, demonstrating the capability of the ML approach to better capture the spatial variability of XCO_2. The extended coverage of ML XCO_2 is expected to enhance our comprehensive understanding of global atmospheric CO_2 distributions. However, it is important to carefully consider the use of ML XCO_2 when analyzing relationships between this ML XCO_2 and its related variables. Since the relationships between XCO_2 and influencing factors are already learned in the model training process, reanalyzing the relationships using ML XCO_2 with co-emitted species (e.g., NO_2, CO) or biospheric indicators (e.g., SIF) may introduce potential errors due to embedded relationships in the training process.

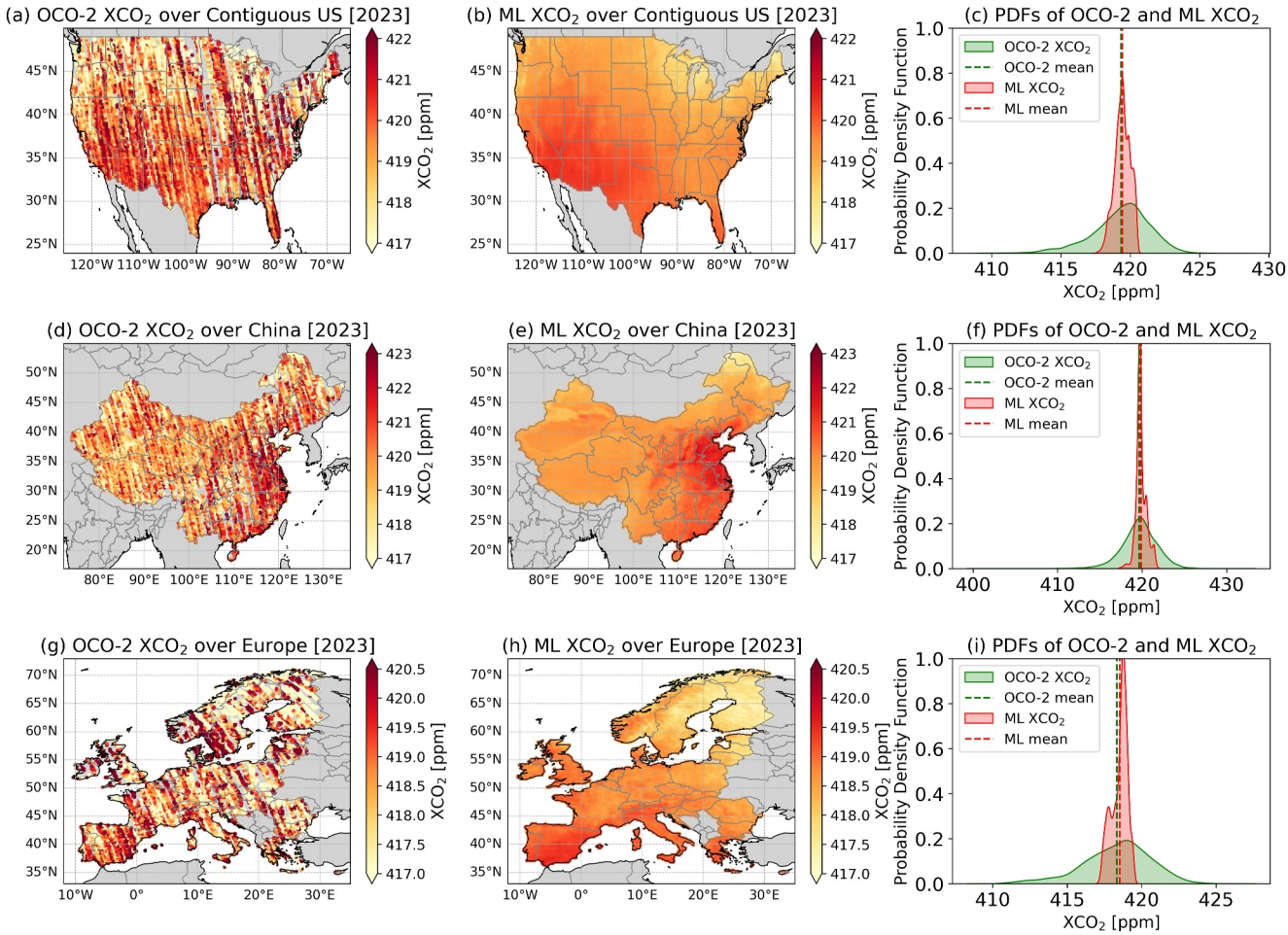


Figure 9. Spatial distributions and probability density functions (PDFs) of XCO_2 over the contiguous United States, China, and Europe in 2023. Panels (a), (d), and (g) show OCO-2 XCO_2 over the contiguous United States, China, and Europe, respectively. Panels (b), (e), and (h) depict ML XCO_2 for the same regions. Panels (c), (f), and (i) present the corresponding PDFs of OCO-2 XCO_2 (green) and ML XCO_2 (red) across the contiguous United States, China, and Europe, respectively. Note that the color bar ranges in the left and center columns are different.

We conducted a feature importance analysis to assess the contributions of variables used in this study (Figure S12 in Supporting Information S1). In the land model (Figure S12a in Supporting Information S1), spatial (latitude, longitude: Lat, Lon) and temporal information (year, day of the year: Year, DOY) are the dominant features. As expected, atmospheric CO_2 exhibits clear seasonal and interannual variation patterns (Figures 4–6 and Figure S5 in Supporting Information S1), and ML XCO_2 effectively captures these spatiotemporal variation patterns. Similar findings were reported in previous studies, where spatiotemporal information played an important role in XCO_2 estimation (C. He et al., 2022; S. He et al., 2023; W. Liu et al., 2024). Other variables exhibit similar levels of feature importance. Among these, CO shows a high feature importance following spatial and temporal variables. This is because CO and CO_2 share common emission sources such as fossil fuel combustion. Atmospheric transport by wind, especially high wind speed (C. Park et al., 2021), also affects CO_2 concentration. NO_2 has a relatively lower contribution in the RF model. One possible explanation is that although NO_2 is co-emitted with CO_2 during the use of fossil fuels, NO_2 has a much shorter lifetime (a few hours) compared to CO_2 (hundreds to thousands of years), which may reduce its influence on XCO_2 variations on a global scale. In addition, because NO_2 also exhibits spatial and temporal variability, the RF model may primarily explain XCO_2 variations using spatial and temporal variables, thereby reducing the relative influence of NO_2. In the ocean model, we found similar feature importance patterns (Figure S12b in Supporting Information S1). CO and NO_2 have relatively lower contributions in the ocean model. This is likely because primary emission sources of NO_2 and CO are located over land, leading to relatively lower contributions of these variables over ocean in the RF model. However, it should be noted that the importance score represents only the relative importance of each input

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variable, and the lower score does not mean that the variable is not necessary in the model. Additionally, feature importance scores may not directly correspond to the physical contributions to estimate independent variable (Long et al., 2022).

This study had certain limitations. ML XCO_2 relies on data input from Sentinel-5P TROPOMI retrievals and ERA5 reanalysis data. Although ERA5 reanalysis data are consistently available across all pixels, the Sentinel-5P TROPOMI retrievals are not always available for every pixel because aerosol and cloud contaminated Sentinel-5P TROPOMI pixels are filtered out. Consequently, our ML XCO_2 estimates could only be obtained in pixels for which the Sentinel-5P TROPOMI products were available (Figure S1 in Supporting Information S1). Although our approach effectively fills the observation gaps in XCO_2 (Figure S11 in Supporting Information S1 and Figure 9), it indicates that our ML XCO_2 estimates may fully resolve the cloud and aerosol contamination issues in OCO-2 XCO_2 retrievals issues, as ML XCO_2 are still absent where Sentinel-5P TROPOMI data are missing. Additionally, the ML XCO_2 period from May 2018 to December 2023 corresponded to the available data set for Sentinel-5P TROPOMI products at the start of our research. This relatively short timeframe limited the ability to assess the long-term characteristics of atmospheric CO_2 (Figures 6 and 8). Despite these limitations, our study effectively demonstrates how an ML approach can be utilized to fill the gaps in XCO_2 data on a global scale and enhance our understanding of the global carbon cycle. Furthermore, the application of our approach over a longer period is expected to provide more comprehensive insight into the long-term dynamics of atmospheric CO_2.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

The OCO-2 data sets (OCO-2/OCO-3 Science Team, Payne & Chatterjee, 2022) are available from NASA Goddard Earth Science Data and Information Service Center (https://daac.gsfc.nasa.gov/). The Sentinel-5P TROPOMI data sets (Copernicus Sentinel data processed by the European Space Agency, 2021a, 2021b; Guanter, Bacour, Schneider, Aben, van, et al., 2021) are available from Sentinel-5P Data Hub (https://dataspace.copernicus.eu/explore-data/data-collections/sentinel-data/sentinel-5p). The ERA5 data sets (Hersbach et al., 2023) are available from Copernicus Climate Data Store (https://cds.climate.copernicus.eu/). The TCCON data sets (Total Carbon Column Observing Network (TCCON) Team, 2022) used in this study are obtained from the TCCON Data Archive hosted by CaltechData (https://tccondata.org).

Acknowledgments

The authors would like to thank the editor and anonymous reviewers for providing valuable comments and suggestions that have significantly improved the quality of the paper, and for their time and effort in reviewing our manuscript. We acknowledge the efforts of NASA and ESA with respect to providing the OCO-2 and Sentinel-5P TROPOMI data products. The authors acknowledge the efforts of Copernicus Climate Change with respect to providing the ERA5 reanalysis data. We gratefully acknowledge TCCON site PIs for the data used in this work. This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (RS-2024-00350480). Also, this work was supported by a grant from the National Institute of Environmental Research (NIER), funded by the Ministry of Environment (ME) of the Republic of Korea (NIER-2023-04-02-211).

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